

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Bhuvaneshwarani J ⁽¹⁹²⁵¹¹⁴⁰²⁾ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:







Annexure A

Constraint-Based Problem Solving

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1. A courier company must assign 3 delivery trucks (Truck1, Truck2, Truck3) to 3 delivery time-windows (9-11 AM, 11 AM-1 PM, 1-3 PM) such that:

Constraints:

- i. Truck1 cannot deliver in the 1-3 PM window
- ii. Truck2 must deliver before Truck3 ($9-11 < 11-1 < 1-3$)
- iii. Only one truck per time-window
- iv. Truck3 cannot deliver in the 9-11 AM window
- v. All trucks must be assigned exactly one time-window

Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

2. A package-delivery robot navigates a 5x5 city-block grid from Start (S) to a delivery address (G). Certain streets are one-way, restricting movement direction in some cells.

Constraints:

- i. Movement allowed: Up, Down, Left, Right, but one-way cells restrict at least one direction
- ii. Cost of each move = 1
- iii. Cannot move through blocked/no-entry cells in the restricted direction
- iv. Use Manhattan Distance heuristic

Using the above constraints and BFS Search Algorithm, show step-by-step node expansion and frontier updates, and determine the optimal path and cost.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

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Courier company must assign 3 delivery trucks
(Truck 1, Truck 2, Truck 3) to 3

* Backtracking Search (Truck scheduling):

⇒ Variables

- Truck 1 (T_1)
- Truck 2 (T_2)
- Truck 3 (T_3)

⇒ Time windows

$$W_1 = 9-11 \text{ AM}$$

$$W_2 = 11 \text{ AM} - 1 \text{ PM}$$

$$W_3 = 1-3 \text{ PM}$$

⇒ Constraints

$$T_1 \neq W_3$$

T_2 before T_3

one truck per window

$$T_3 \neq W_1$$

Every truck gets exactly one window

⇒ step by step - Backtracking

Step 1

Assign $T_1 \rightarrow W_1$

current

$$T_1 = W_1$$

Step 2

Try $T_2 \rightarrow W_1$

constraints violated

- only one truck per window

Backtrack

Try $T_2 \rightarrow W_2$

current

$$T_1 = W_1$$

$$T_2 = W_2$$

Step 3

Try $T_3 \rightarrow w_1$

Constraint violated

T_3 cannot be in w_1

Back track:

Try $T_3 \rightarrow w_2$

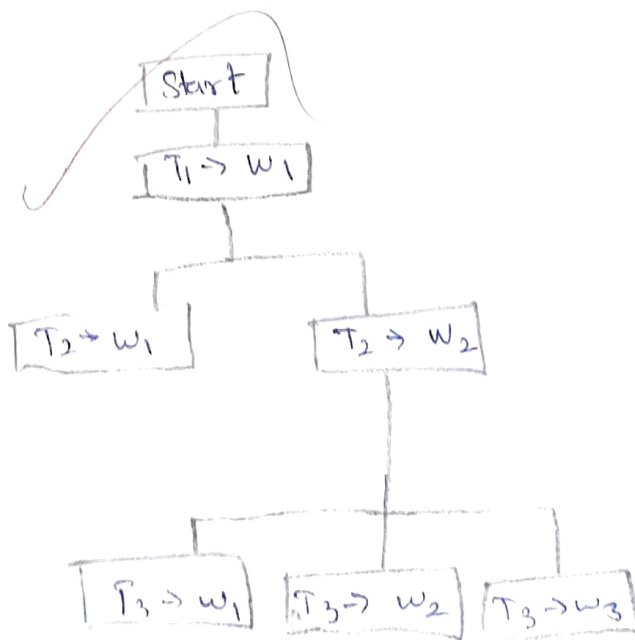
constraints violated

only one truck per window

Back Track

Try $T_3 \rightarrow w_3$

Search Tree



→ package - delivery robot navigates a 5x5 grid from start (s) to a delivery.

BFS search (5x5) grid

Grid:

```
S . . x .  
 . x . x .  
 . . . . .  
 x x . x .  
 . . . . G
```

Legend

S = start x = Blocked

G = goal . = Open

Manhattan Distance

$$|x_1 - x_2| + |y_1 - y_2|$$

BFS node Expansion

Step 1:

Expand
S

Frontier:

(0,1), (1,0)

Step 2:

Expand
(0,1)

Frontier:

(1,0), (0,2)

Step 3:

Expand

$(1,0)$

Frontier:

$(0,2), (2,0)$

Step 4:

Expand

$(0,2)$

Frontier:

$(2,0), (1,2)$

Step 5:

Expand

$(2,0)$

Frontier

$(1,2), (2,1)$

Step 6:

Expand

$(1,2)$

Frontier

$(2,1), (2,2)$

Total cost

Each move costs 1

Number of moves = 8

optimal cost = 8

BFS:



explores nodes level by level and always finds the shortest path when all moves costs are equal.